Teaching Statement

My introduction to teaching computer science began during an undergraduate summer research program at Princeton University, when I taught a lecture on approximation algorithms[1] while the instructor was traveling. It took me a whole week, and many practice runs, but the process of preparing to teach complex concepts in a creative way, showed me how much care and effort is required to effectively teach—but also how much I enjoyed it. Since then, I have taken advantage of different teaching opportunities during my PhD, and as an industry researcher through:

- Running a summer course: In the final year of my PhD, I ran an undergraduate research summer course at Brown. This involved selecting applicants from a large candidate pool, designing course materials and TAs duties, interacting with administrators for room assignments and bookings, holding office hours, and preparing materials for lectures, assignments, and research programs.
- Summer schools, guest lectures & seminars: I've been invited to guest lecture on specific course materials at Brown University, UC Berkeley, and to speak at university seminars at Pomona University, Rutgers University, and Princeton University. I've also taught a series of safety and language modeling lectures at the M2L summer school in 2022 and 2024.
- Advising students & teaching researchers: As an industry researcher, I've advised 6 PhD students during internships at DeepMind. I mentored several undergrads, as well as younger grad students during my PhD. I taught tutorials to non-language researchers at DeepMind: when the company pivoted towards LLM products, I was asked to do a 1-week tutorial on training language models for the multi-agent and game theory team.
- Advisory boards & academic engagement: I was invited to be part of an academic advisory board at Brown University, to help shape future CS courses in light of evolving AI tools in education. I have also co-taught tutorials on language models to academic and industry communities.

My philosophy towards teaching and mentorship has been significantly shaped by the above experiences. Through these, my teaching and mentoring strategy has centered on three principles: encouraging critical thinking, building intuition through concrete examples, and being able to adapt to each student. As a professor, I hope to bring these principles to both classroom instruction and research advising.

Teaching & Lecturing

One of the things that I quickly learned is that students learn best when encouraged to do two things: first, to focus on the fundamental concepts rather than details of complex methods; and second, to explicitly question the assumptions, limitations and mechanisms behind them. The first is even more important today in this fast-changing environment of new methodologies; and the second is even more crucial when students use AI assistant tools whose help, when used incorrectly, might actually hinder, rather than aid student's learning and developmental processes.

Through my career, both in graduate school and industry, I've had the opportunity to teach **large scale tutorials** attended by upto 1000 virtual participants, **summer school lectures** attended by upto 300 in-person students, **college-level courses** attended by undergraduate and graduate students, and **academic and industry tutorials** attended by hundreds of post-graduate researchers across disciplines. These audiences have varied in their backgrounds, age groups, disciplines, and research interests; requiring a balance of teaching conceptual intuition along with necessary implementation details, while ensuring that every person's academic needs are fulfilled.

Moreover, being on the **academic advisory board** at Brown University has furthered my motivation to be a part of the academic teaching ecosystem. In this era of AI assistants that make traditional teaching and homework assignment regimes difficult, I'm excited to find new ways to engage, challenge and evaluate students. Rather than restricting the usage of AI assistants, I'm interested to come up with creative ways of designing assignments that require demonstrations of independent thinking, critical evaluations of model-generated solutions, and to teach them

[1] We had been doing research on this topic that whole summer. The semester after, I discovered research in natural language processing (NLP) and never went back to theoretical computer science.

to use automated systems while recognizing their failures. I think this is a hard problem, but a necessary, and even exciting one for us to tackle; and I view this as an opportunity to rethink education techniques while preserving the institutional functions of academia.

Mentoring & Advising

Mentoring has been among the most fulfilling parts of my job at DeepMind. As a senior researcher, I've been fortunate to have directly or jointly **advised 6 PhD students** through various internships. These collaborations have led to several first-author research papers from respective students I mentored; and have shaped my understanding of what effective research advising requires. I've also been an active part of Google's internal career mentorship program, having **mentored 11 engineers and researchers** on career development and industry jobs. Being in these roles has shaped my approach to advising, teaching and mentoring over the years.

First, I've learned to provide **substantial structure early on** in a project, while making sure to **encourage and reward intellectual ownership** and creativity as the project progresses. My favourite part of any research project is when a student proposes a fundamentally different approach from what I had envisioned; taking the project in an unexpected, but more creative direction to answer the research question at hand. Second, I try to **adapt teaching strategies to individual students**. Both in large-scale lectures, as well as individual interactions, being aware of a student's comfort level with the concepts at hand is helpful. I try to tailor my approach to each student differently; some might require many one-to-one interactions to grasp the concept at hand; while others may require only higher-level feedback, and might suffer under the pressure of too many interactions. Learning from these experiences with students I have taught and advised has been immensely helpful as a **generalization strategy**: seeing what challenges have been faced by a subset of past students shapes my understanding of what to pay attention to with future students. I'm excited to continually learn to adapt to what each student needs, in order to deliver the education they need from me. Lastly, I believe that an academic environment can only function well when **students feel comfortable debating, disagreeing and challenging** what others say. I aim to create environments where junior researchers feel comfortable questioning assumptions, proposing alternatives, and taking ownership of ideas that are fundamentally different from what currently exists.

Future Teaching

I'm excited to teach both types of classes: general, **introductory computer science classes** (e.g., artificial intelligence or machine learning) and in-depth, **graduate level seminars on emerging topics** (e.g., modeling safety risks or longitudinal data collection). I am eager to teach courses that my research and expertise span i.e., **natural language processing, reinforcement learning, deep learning, safety and ethics, alignment**. Separately, I'm very excited to formulate **interdisciplinary courses** e.g., examining the relationship between AI and society, human-AI interactions, large-scale data collections, social impacts and foresight about increasingly capable AI systems. Mentoring and teaching are such a fundamental part of my motivation to continue an academic career, and I'm excited to implement new and creative ways to help students learn in this rapidly evolving landscape.

As a faculty member, I plan to build a **collaborative**, **intellectually creative**, and **interdisciplinary research group** where students feel empowered to pursue ambitious ideas at the intersection of safety, AI, and language technologies. I plan to combine regular individual mentoring with group meetings, reading groups and collaborative events that bring opportunities for students to learn from one another. My goal is to not just teach students how to do develop skills to do successful research, but to develop the independence, confidence and judgement to **recognize important and ambitious research directions**. Through my teaching, mentoring, and academic service, I'm excited to create an environment where students learn, teach, and challenge ideas that lead to creative and rigorous research that will shape the future of AI.